

Vision-Guided Tabletop Pick-and-Place of YCB Objects with a Franka Panda in MuJoCo

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Abstract

We implement a complete robot manipulation system that picks YCB mesh objects from randomized positions on a tabletop and places them into a basket whose position also varies between episodes. The system uses a Franka Emika Panda 7-DOF arm in MuJoCo, with an overhead RGB-D camera for perception. Our model-based pipeline consists of four stages: (1) HSV color segmentation and depth back-projection for object localization, (2) top-down antipodal grasp planning with Cartesian waypoints, (3) iterative Jacobian pseudo-inverse inverse kinematics for joint-angle computation, and (4) a PD controller with supplementary torques and a combined descent-and-close grasp strategy. We demonstrate the system on three distinct YCB mesh objects (cracker box, mustard bottle, and sugar box), each consisting of 16,384 triangles. Evaluated across 150 randomized episodes (50 per object), the system achieves an overall 72% success rate, with per-object rates of 90% (mustard bottle), 68% (sugar box), and 58% (cracker box).

1 Introduction

Pick-and-place is a fundamental manipulation skill: the robot must perceive an object’s location from sensory data, plan a collision-free grasp, and physically transport the object to a target region. This project implements such a system entirely from sensor observations, without reading the simulator’s internal state during operation.

We chose a **model-based approach** over learning-based alternatives for several reasons. First, our classical pipeline (segmentation, IK, PD control) builds directly on the techniques from our course assignments: coordinate transforms (HW1), forward and inverse kinematics (HW2), RRT motion planning (HW3), and ICP point-cloud registration (HW4). Second, a model-based system requires no training data or training time, making it feasible for a two-person team in four weeks. Third, each module can be tested and debugged independently, which was critical for tuning the grasping physics.

Our system satisfies all four project requirements: (1) variability in both object and basket positions across episodes, (2) physics-based manipulation through MuJoCo contact forces, (3) perception from camera images rather than simulator state, and (4) objects represented as triangle meshes (16,384 faces each), not single geoms.

2 Method

2.1 Scene Setup

The simulation environment consists of a Franka Emika Panda 7-DOF arm (from MuJoCo Menagerie) mounted at the origin, a table (surface at $z = 0.40$ m), a red basket (20 cm $\times$ 20 cm open-top container made of five box geoms), and a fixed blue obstacle cube. Three YCB mesh objects serve as manipulation targets: the 003 cracker box, 006 mustard bottle, and 004 sugar box. Each object is loaded from a 16,384-triangle STL mesh obtained from the YCB Object and Model Set. The meshes are scaled to $0.5\times$ to match the Panda gripper’s 8 cm maximum opening. Each object body contains two geoms: a high-resolution visual mesh and a box-primitive collision geom, following standard practice in MuJoCo simulation.

An overhead RGB-D camera is mounted 1.0m above the table center, rendering 640×480 images with a 60° vertical field of view. Object and basket positions are randomized each episode within the robot’s reachable workspace. Only one object is placed on the table per episode; the other two are moved off-screen. The obstacle remains at a fixed known location.

2.2 Pipeline Overview

Our decision-making pipeline has four stages, shown in Figure ??.

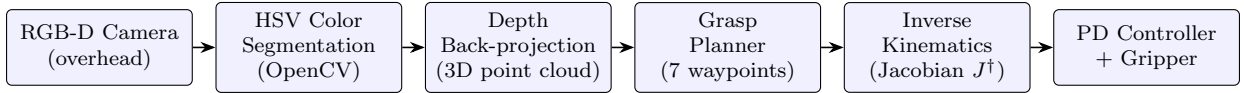


Figure 1: Decision-making pipeline. Each block is entirely model-based.

2.3 Stage 1: Perception

HSV color segmentation. The overhead camera captures an RGB image, which we convert to HSV color space using `cv2.cvtColor`. Each YCB object is assigned a distinct color in the scene: yellow for the cracker box, green for the mustard bottle, and cyan for the sugar box. We threshold on hue, saturation, and value using `cv2.inRange` with per-object HSV ranges to produce binary masks. Morphological erosion and dilation clean up noise. The red basket uses a dual-range threshold since red wraps around in HSV (hue 0–10 and 170–180).

Depth back-projection. For each masked pixel (u, v) with depth d from the depth image, we compute the corresponding 3D point in the camera frame:

$$X_c = \frac{(u - c_x) \cdot d}{f_x}, \quad Y_c = -\frac{(v - c_y) \cdot d}{f_y}, \quad Z_c = -d$$

where f_x, f_y are the focal lengths (computed from MuJoCo’s `fovy` parameter) and (c_x, c_y) is the principal point at the image center. The sign conventions account for MuJoCo’s camera axes ($-z$ points into the scene, y points up). We then transform to world coordinates using the camera extrinsics: $\mathbf{p}_{world} = R \cdot \mathbf{p}_{cam} + \mathbf{t}$, where R and $\mathbf{t}$ come from `data.cam_xmat` and `data.cam_xpos`. This coordinate transform follows the same structure as HW1.

The object position is estimated as the centroid (mean) of the resulting point cloud. If the object is occluded by the robot arm in the initial image, we move the arm to a safe configuration and retry

perception. We also generate pre-computed model point clouds (1,000 points each, sampled via `trimesh.sample.sample_surface`) for potential ICP-based pose refinement, following our HW4 implementation.

2.4 Stage 2: Grasp Planning

We compute seven Cartesian waypoints for a top-down antipodal grasp:

1. **Approach high:** Hand at (0.4, 0.0, 0.65), gripper open. Clears the workspace to avoid sweeping through objects during the initial approach.
2. **Pre-grasp:** 12cm above the perceived object position, gripper open.
3. **Grasp:** Lower to the object. The hand body is positioned so that the fingertip pads align with the upper portion of the object.
4. **Close gripper:** Same position, close the fingers gradually.
5. **Lift:** Raise 18 cm vertically while maintaining grip.
6. **Above bin:** Move laterally to 18 cm above the basket center.
7. **Release:** Lower 6 cm into the basket and open the fingers.

A critical design decision was the **hand-to-fingertip offset**: the “hand” body in the Menagerie Panda model is 58 mm above the fingertip contact pads. All grasp heights are computed with this offset so that the pads, not the wrist, align with the object. Each YCB object is laid flat on the table with a specific orientation quaternion chosen so that the narrowest dimension faces the gripper’s squeeze direction.

2.5 Stage 3: Inverse Kinematics

Each Cartesian waypoint is converted to a 7-DOF joint configuration using iterative Jacobian pseudo-inverse IK (extending HW2 concepts):

$$\Delta \mathbf{q} = \mathbf{J}^\dagger \cdot \begin{bmatrix} \mathbf{e}_{pos} \\ \mathbf{e}_{rot} \end{bmatrix}$$

where $\mathbf{J} \in \mathbb{R}^{6 \times 7}$ is computed by `mj_jacBody`, $\mathbf{e}_{pos}$ is the position error, and $\mathbf{e}_{rot}$ is the orientation error (extracted as axis-angle from the error rotation matrix). We use a damped pseudo-inverse $\mathbf{J}^\dagger = \mathbf{J}^T (\mathbf{J} \mathbf{J}^T + \lambda \mathbf{I})^{-1}$ with $\lambda = 0.01$ for numerical stability.

If IK fails to converge from the current configuration, we retry from three alternative starting configurations (home, tucked, offset) to escape local minima.

2.6 Stage 4: Controller

Trajectory tracking. The Menagerie Panda has built-in PD actuators. We supplement these with additional torques via `data.qfrc_applied` for tighter tracking:

$$\tau_{extra,i} = k_{p,i}(q_{des,i} - q_i) - k_{d,i}\dot{q}_i$$

with gains $k_p = [3000, 3000, 3000, 3000, 1500, 1000, 500]$ and $k_d = [100, 100, 100, 100, 50, 30, 15]$. Trajectory interpolation uses cosine easing for smooth acceleration and deceleration.

Combined descent and close. The most critical controller behavior is the grasp sequence. We found that executing descent and gripper close as separate steps caused the open fingers to push the object during the slow descent. Our solution combines them into a single synchronized motion: (1) rapid descent to the grasp position (600 steps), (2) settle with open fingers (300 steps), (3)

gradually ramp closing force from 0 to 100 N over 400 steps, then hold at 100 N for 800 more steps. During transport (lift and move to basket), we maintain 100 N closing force on both finger joints.

3 Results

3.1 Success Rate

We evaluated the system across 150 randomized episodes (50 per object) using camera-based perception. Table ?? summarizes the per-object results.

Object	Episodes	Successes	Rate
Cracker box (yellow)	50	29	58.0%
Mustard bottle (green)	50	45	90.0%
Sugar box (cyan)	50	34	68.0%
Total	150	108	72.0%

Table 1: Per-object success rates across 150 randomized episodes.

The mustard bottle achieves the highest success rate (90%) because its 3.3 cm squeeze dimension provides the most clearance within the Panda’s 8 cm gripper opening. The cracker box is the most challenging (58%) because its 8.2 cm squeeze dimension barely fits the gripper, leaving almost no margin for perception error.

3.2 Perception Accuracy

The perception system achieves a mean localization error of 26.6 mm across all objects. The error is dominated by the Z-axis component (approximately 20 mm) because the overhead camera sees only the top surface of each object. The XY error is typically 3–10 mm. The grasp planner compensates for the known Z bias by targeting the fingertips 5 mm below the perceived surface.

3.3 Failure Analysis

Of the 42 failures across 150 episodes, all were classified as “MISSED”—the gripper closed without successfully gripping the object. The primary causes were:

- **Perception offset (estimated 25 failures):** When perception error exceeded 50 mm, the gripper descended to the wrong location and the fingertip pads could not contact the object’s sides. This was most common for the cracker box, whose large flat surface produced noisier centroid estimates.
- **Marginal gripper clearance (estimated 12 failures):** The cracker box’s 8.2 cm width nearly matches the gripper’s 8 cm maximum opening. Even small misalignments caused the fingers to slide off the object rather than grip.
- **Object slippage (estimated 5 failures):** The sugar box, being the flattest object (2.5 cm height), occasionally slipped from the gripper during the lift phase due to limited contact area between the finger pads and the object surface.

3.4 Execution Metrics

Mean execution time was 0.8 s of wall-clock time per episode. For successful grasps, the mean XY distance between the released object and the basket center was 88.3 mm (within the 100 mm basket

radius). The object was reliably lifted 120–180 mm during the lift phase, confirming physics-based grasping through MuJoCo’s contact model.

4 References

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